

1. Dense pixel backbone

AlphaEarth
64, 256, 256



Tessera
128, 256, 256



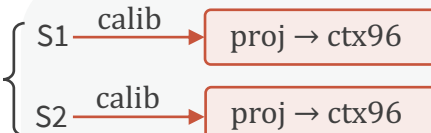
Spatial Gate^[1]

$gF_{\alpha} + (1 - g)F_{\text{t}}$

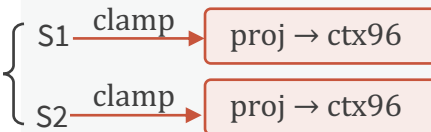
F_{pixel}

2. Coarse token conditioning

Terramind
768, 16, 16



Thor
768, 16, 16



**Cross-source
Self-attention**
(zero-init)

3. Fusion

$$F_{\text{out}} = F_{\text{pixel}} + \sum_i \delta_i$$

δ_i

FiLM^[2] γ_i, β_i
+ add A_i
+ gate $\sigma(g_i)$
up $16^2 \rightarrow 256^2$

x4 one per source

4. Split-trunk multi-task head

S_{seg}

Seg trunk

Presence head
bld/veg/water

ch0-2

presence
gate

S_{hgt}

Height trunk

Height head

ch3

Output

Two-stage training:

stage 1 coupled ($S_{\text{seg}} = S_{\text{hgt}} = 1$)

stage 2 purify:

- seg-purify (+ cIDice) ($S_{\text{hgt}} = 0$) $\Rightarrow$ ch 0-2
- height-purify ($S_{\text{seg}} = 0$) $\Rightarrow$ ch 3